Creating and Customizing Rails Generators & Templates

Rails generators are an essential tool if you plan to improve your workflow. With this guide you will learn how to create generators and customize existing ones.

1 First Contact

When you create an application using the `rails` command, you are in fact using a Rails generator. After that, you can get a list of all available generators by just invoking `rails generate`:

```
$ rails new myapp
$ cd myapp
$ bin/rails generate
```

You will get a list of all generators that comes with Rails. If you need a detailed description of the helper generator, for example, you can simply do:

```
$ bin/rails generate helper --help
```

2 Creating Your First Generator

Since Rails 3.0, generators are built on top of Thor. Thor provides powerful options for parsing and a great API for manipulating files. For instance, let's build a generator that creates an initializer file named `initializer.rb` inside `config/initializers`.

The first step is to create a file at `lib/generators/initializer_generator.rb` with the following content:

```
class InitializerGenerator < Rails::Generators::Base
  def create_initializer_file
    create_file "config/initializers/initializer.rb", "# Add
initialization content here"
  end
end
```

`create_file` is a method provided by `Thor::Actions`. Documentation for `create_file` and other Thor methods can be found in Thor's documentation

Our new generator is quite simple: it inherits from `Rails::Generators::Base` and has one method definition. When a generator is invoked, each public method in the generator is executed sequentially in the order that it is defined. Finally, we invoke the `create_file` method that will create a file at the given destination with the given content. If you are familiar with the Rails Application Templates API, you'll feel right at home with the new generators API.

To invoke our new generator, we just need to do:

```
$ bin/rails generate initializer
```

Before we go on, let's see our brand new generator description:

```
$ bin/rails generate initializer --help
```

Rails is usually able to generate good descriptions if a generator is namespaced, as `ActiveRecord::Generators::ModelGenerator`, but not in this particular case. We can solve this problem in two ways. The first one is calling `desc` inside our generator:

```
class InitializerGenerator < Rails::Generators::Base
```